

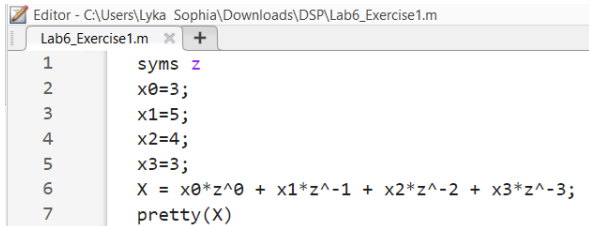
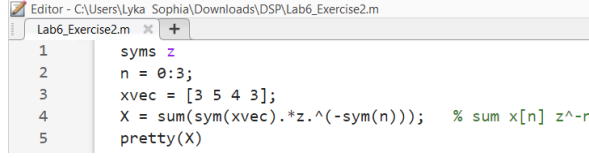
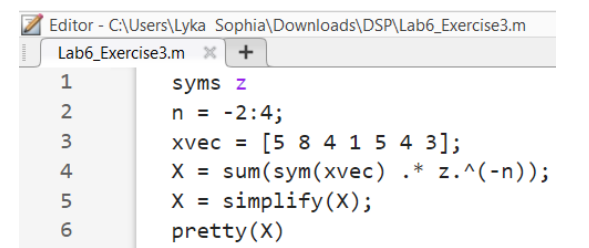
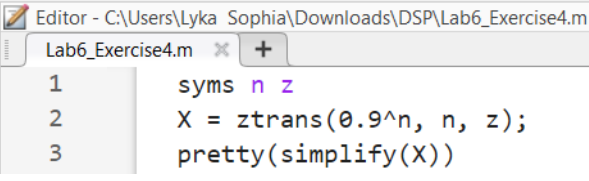
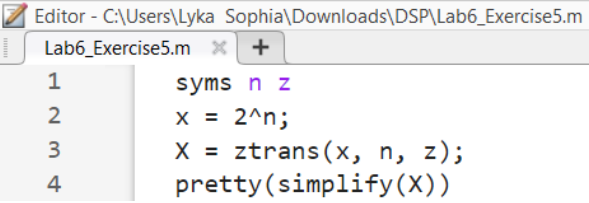


Name:	Lyka Sophia S. Jardeleza	Date:	10/31/2025
Program:	BSCpE	Section:	A461

A. Guided Exercises:

(Please include the following: 1. Photos of the codes and 2. Outputs after executing the MATLAB code.)

Exercises

Photo/s of MATLAB output	Photos of MATLAB Codes
1) <pre>>> Lab6_Exercise1</pre> $\frac{5z^4 + 4z^3 - z^2 + 3z}{z^2}$	1)  <pre>1 syms z 2 x0=3; 3 x1=5; 4 x2=4; 5 x3=3; 6 X = x0*z^0 + x1*z^-1 + x2*z^-2 + x3*z^-3; 7 pretty(X)</pre>
2) <pre>>> Lab6_Exercise2</pre> $\frac{5z^4 + 4z^3 - z^2 + 3z}{z^2}$	2)  <pre>1 syms z 2 n = 0:3; 3 xvec = [3 5 4 3]; 4 X = sum(sym(xvec). * z.^(-sym(n))); % sum x[n] z^-n 5 pretty(X)</pre>
3) <pre>>> Lab6_Exercise3</pre> $\frac{5z^6 + 8z^5 + 4z^4 + z^3 + 5z^2 + 4z + 3}{z^4}$	3)  <pre>1 syms z 2 n = -2:4; 3 xvec = [5 8 4 1 5 4 3]; 4 X = sum(sym(xvec) .* z.^(-n)); 5 X = simplify(X); 6 pretty(X)</pre>
4) <pre>>> Lab6_Exercise4</pre> $\frac{z^9}{z^{10}}$	4)  <pre>1 syms n z 2 X = ztrans(0.9^n, n, z); 3 pretty(simplify(X))</pre>
5) <pre>>> Lab6_Exercise5</pre> $\frac{z}{z^2}$	5)  <pre>1 syms n z 2 x = 2^n; 3 X = ztrans(x, n, z); 4 pretty(simplify(X))</pre>



6)

```
>> Lab6_Exercise6
n
2
```

7)

```
>> Lab6_Exercise7

ro =

-4.0000
-2.0000
 1.0000

c1 =

1/2

c2 =

1/6

c3 =

1/3

      1      1      1
----- + ----- + -----
3 (z - 1)  6 (z + 2)  2 (z + 4)
```

6)

```
Editor - C:\Users\Lyka_Sophia\Downloads\DSP\Lab6_Exercise6.m
Lab6_Exercise6.m x +
1      syms n z
2      Xz = z/(z-2);
3      x_n = iztrans(Xz, z, n);
4      pretty(simplify(x_n))
```

7)

```
Editor - C:\Users\Lyka_Sophia\Downloads\DSP\Lab6_Exercise7.m
Lab6_Exercise7.m x +
1      % Exercise 7: Partial Fraction Expansion
2      syms z
3
4      % Denominator coefficients (z^3 + 5z^2 + 2z - 8)
5      A = [1 5 2 -8];
6
7      % Find poles (roots of denominator)
8      ro = roots(A)
9
10     % Define X(z)
11     X = (z^2 + 3*z + 1) / (z^3 + 5*z^2 + 2*z - 8);
12
13     % Compute residues (c1, c2, c3) using the limit formula
14     c1 = limit((z - ro(1)) * X, z, ro(1))
15     c2 = limit((z - ro(2)) * X, z, ro(2))
16     c3 = limit((z - ro(3)) * X, z, ro(3))
17
18     % Optional: Display the partial fraction expansion
19     pretty(simplify(c1/(z - ro(1)) + c2/(z - ro(2)) + c3/(z - ro(3))))
```

B. Problem Exercises:

(Please include the following: 1. Photos of the codes and 2. Outputs after executing the MATLAB code.)

Determine the relationship (linearity and time-invariance relationship) of the given systems in the Lab Guide.

1. Problem 1

Photo/s of MATLAB output	Photos of MATLAB Codes
a) <pre>>> Lab6_Problem1 --- Problem 1 (a): Z-Transform of x[n] = n^2 u[n] --- z (z + 1) ----- 3 (z - 1)</pre>	a) <pre>Editor - C:\Users\Lyka_Sophia\Downloads\DSP\Lab6_Problem1.m Lab6_Problem1.m x + 1 % (a) Compute the Z-transform 2 syms n z 3 x = n^2; 4 X = ztrans(x, n, z); 5 6 disp('--- Problem 1 (a): Z-Transform of x[n] = n^2 u[n] ---') 7 pretty(simplify(X))</pre>



b)

```
--- Problem 1 (b): Inverse Z-Transform ---
      / n - 1 \
3 n + 2 |      | - 2
      \  2    /
```

b)

```
Editor - C:\Users\Lyka_Sophia\Downloads\DSP\Lab6_Problem1.m
Lab6_Problem1.m
9      % (b) Verify by inverse Z-transform
10     disp(' ')
11     disp('--- Problem 1 (b): Inverse Z-Transform ---')
12     x_reconstructed = iztrans(X, z, n);
13     pretty(simplify(x_reconstructed))
```

2. Problem 2

Photo/s of MATLAB output	Photos of MATLAB Codes
a) <pre>>> Lab6_Problem2 --- Problem 2 (a): Z-Transform of f1[n] = [-3, 5, 6, 7, 8], -2 <= n <= 2 --- 4 3 2 - 3 z + 5 z + 6 z + 7 z + 8 ----- 2 z</pre> b) <pre>--- Problem 2 (b): Z-Transform of f2[n] = [-3, 5, 6, 7, 8], 0 <= n <= 4 --- 4 3 2 - 3 z + 5 z + 6 z + 7 z + 8 ----- 4 z</pre>	a) <pre>Editor - C:\Users\Lyka_Sophia\Downloads\DSP\Lab6_Problem2.m Lab6_Problem2.m 1 % (a) Compute the Z-transform of f1[n] = [-3, 5, 6, 7, 8], -2 <= n <= 2 2 syms z 3 n1 = -2:2; 4 f1 = [-3 5 6 7 8]; 5 6 disp('--- Problem 2 (a): Z-Transform of f1[n] = [-3, 5, 6, 7, 8], -2 <= n <= 2 ---') 7 Xf1 = sum(sym(f1) .* z.^(-sym(n1))); 8 pretty(simplify(Xf1))</pre> b) <pre>Editor - C:\Users\Lyka_Sophia\Downloads\DSP\Lab6_Problem2.m Lab6_Problem2.m 1 % (b) Compute the Z-transform of f2[n] = [-3, 5, 6, 7, 8], 0 <= n <= 4 2 n2 = 0:4; 3 f2 = [-3 5 6 7 8]; 4 5 disp('--- Problem 2 (b): Z-Transform of f2[n] = [-3, 5, 6, 7, 8], 0 <= n <= 4 ---') 6 Xf2 = sum(sym(f2) .* z.^(-sym(n2))); 7 pretty(simplify(Xf2))</pre>

3. Problem 3

Photo/s of MATLAB output	Photos of MATLAB Codes
a) <pre>>> Lab6_Problem3 --- Problem 3 (a): Partial-Fraction Expansion of X(z) = (z^3 + 3z + 2) / (2z^2 + z - 1) --- 2 29 1 z + ----- + ----- 2 3 (z + 1) (2z - 1) 12 4 Verification of Problem 3 (a): Simplify back to original X(z) ans = (z^3 + 3*z + 2)/(2*z^2 + z - 1)</pre> b) <pre>--- Problem 3 (b): Partial-Fraction Expansion of X(z) = (2z^2 + z - 1) / (z^3 - 3z + 2) --- 13 2 5 9 (z - 1) + ----- + ----- 3 (z - 1) 9 (z + 2) Verification of Problem 3 (b): Simplify back to original X(z) ans = (2*z^2 + z - 1)/((z - 1)^2*(z + 2))</pre>	a) <pre>Editor - C:\Users\Lyka_Sophia\Downloads\DSP\Lab6_Problem3.m Lab6_Problem3.m 1 % (a) Compute the partial-fraction expansion of 2 % X(z) = (z^3 + 3z + 2) / (2z^2 + z - 1) 3 syms z 4 Xa = (z^3 + 3*z + 2) / (2*z^2 + z - 1); 5 6 disp('--- Problem 3 (a): Partial-Fraction Expansion of X(z) = (z^3 + 3z + 2) / (2z^2 + z - 1) ---') 7 Pfa = partfrac(Xa, z); 8 pretty(Pfa) 9 10 disp(' ') 11 disp('Verification of Problem 3 (a): Simplify back to original X(z)') 12 rat_a = simplify(Pfa); 13 [numA, denA] = numden(rat_a); 14 simplify(numA/denA)</pre> b) <pre>Editor - C:\Users\Lyka_Sophia\Downloads\DSP\Lab6_Problem3.m Lab6_Problem3.m 15 % (b) Compute the partial-fraction expansion of 16 % X(z) = (2z^2 + z - 1) / (z^3 - 3z + 2) 17 disp('--- Problem 3 (b): Partial-Fraction Expansion of X(z) = (2z^2 + z - 1) / (z^3 - 3z + 2) ---') 18 Xb = (2*z^2 + z - 1) / (z^3 - 3*z + 2); 19 Pfb = partfrac(Xb, z); 20 pretty(Pfb) 21 22 disp(' ') 23 disp('Verification of Problem 3 (b): Simplify back to original X(z)') 24 rat_b = simplify(Pfb); 25 [numB, denB] = numden(rat_b); 26 simplify(numB/denB)</pre>



z-Transform

C. Findings ,Observations, and Comments:

Guiding Questions :

1. *What did you learn in the guided exercises? Did you learn something new?*
2. *What are your findings, observations, and comments in the guided exercises?*
3. *What are your findings, observations, and comments in the problem exercises?*

In the guided exercises, I learned how to apply the Z-transform and its inverse to different discrete-time signals using MATLAB. I became more familiar with using commands like `ztrans` and `iztrans`, and how each one helps in analyzing and simplifying signals in the z-domain. I also understood better how to express signals as rational functions of z and how the region of convergence affects the validity of the transform. Overall, I learned not just the mathematical process but also how MATLAB automates and verifies these steps, which made the concepts clearer.

From the guided exercises, I observed that MATLAB provides an efficient and accurate way to perform Z-transforms, inverse Z-transforms, and partial-fraction expansions. The exercises showed how theoretical formulas translate directly into symbolic results in MATLAB, confirming that the computations are correct. It also helped me see the relationship between time-domain sequences and their corresponding z-domain representations.

In the problem exercises, I found that applying what I learned in the guided part helped me handle more complex expressions confidently. The problems reinforced my understanding of how to represent signals using Z-transforms and how to check their correctness by using inverse transforms or simplifications. I also observed that partial-fraction expansion is useful for breaking down complicated rational functions into simpler terms, which makes inverse transforms easier. Overall, the problem exercises helped me connect theory to practice and strengthened my understanding of discrete-time signal analysis.